

Explainable AI Framework for Water Potability Prediction Using Machine Learning

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Abstract—Access to safe drinking water is still a significant global public health issue. This study introduces an Explainable AI-based Early Warning Framework that predicts water potability using machine learning methods. The framework covers missing value filling, feature scaling, SMOTE-based class imbalance correction, and supervised classification using Logistic Regression, Decision Tree, and Random Forest models on a dataset of 3,276 water samples, which have nine physicochemical parameters. SHAP explainability analysis was used to enhance model transparency and pinpoint key predictive features. Experimental results indicate that the Random Forest classifier achieved the highest performance, with 99.68% accuracy and 99.67% accuracy in five-fold cross-validation. SHAP analysis showed that conductivity, total dissolved solids (TDS), and trihalomethanes are the most important predictors of water potability. The proposed framework demonstrates the effectiveness of interpretable machine learning for scalable and dependable water quality monitoring applications.

Index Terms—Water Potability, Explainable AI, Random Forest, SHAP, Machine Learning, SMOTE, Water Quality Prediction, Early Warning System

I. INTRODUCTION

The accessibility of pure and potable water is probably one of the most important challenges related to public health and sustainable development of our planet, particularly in rural areas. Contaminated drinking water is considered as a source of disease caused by water pollution that affects human health, sanitation, and overall well-being [1].

Classical methods of water quality assessment are based on laboratory and visual inspection tests. Even though they provide accurate results for chemical composition analysis, they may be regarded as inefficient because of high costs and resource intensiveness at large scales [6], [15].

Therefore, intelligent systems capable of interpreting and predicting water quality are required for addressing this issue. Recent advances in AI and ML demonstrated significant potential for environmental prediction and monitoring tasks. Predictive models based on machine learning efficiently identify complex dependencies between multiple physical/chemical properties of water quality and predict its potability state [2], [14]. Nevertheless, modern machine learning-based models are non-interpretable and black-box ones [9].

The purpose of this paper is to close the identified research gap through the presentation of an *Explainable AI Framework for Water Potability Prediction Using Machine Learning*,

including the following steps: data pre-processing, handling missing data, normalization, SMOTE class imbalance technique, ensemble machine learning classification, and SHAP model explainability. The main contributions of this research include the following:

- A comprehensive, end-to-end ML pipeline for binary water potability classification evaluated on a large-scale dataset.
- Integration of SMOTE to handle severe class imbalance in real-world water quality datasets.
- Application of SHAP explainability to identify key physicochemical drivers of water contamination risk.
- Comparative evaluation of three classifiers using accuracy, precision, recall, F1-score, and five-fold cross-validation.

II. LITERATURE REVIEW

Several studies have explored supervised machine learning techniques for predicting water potability and contamination levels using physicochemical water parameters. Classical models such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), and Artificial Neural Networks (ANN) have demonstrated promising predictive capabilities across a variety of water quality datasets [3], [7], [13].

Feed-forward neural network (FFNN) and adaptive neuro-fuzzy inference system (ANFIS) approaches have been employed for Water Quality Index (WQI) classification, with FFNN demonstrating highly competitive classification performance and ANFIS achieving a regression coefficient of 96.17% [1]. Deep learning approaches such as LSTM, CNN, and hybrid CNN-LSTM architectures have also been utilised for temporal water quality forecasting; however, these methods generally require large-scale temporal datasets and substantial computational resources [4], [12].

Grid search and hyperparameter optimisation strategies have been applied to improve the predictive accuracy of ML models for water quality assessment [8]. Comprehensive reviews of machine learning approaches for water quality prediction further confirm the superiority of ensemble methods over single classifiers across diverse datasets [5].

Explanation of artificial intelligence (XAI) is yet another emerging trend that has been observed in environmental ar-

tificial intelligence systems. Methods of SHAP and feature importance have become popular in recent times as means to improve the interpretability of water resources prediction models through machine learning [9], [16]. While highly accurate prediction results are obtained through the use of these models in the literature, some issues remain, namely model interpretability issues, inadequate handling of the class imbalance issue, and the absence of an early warning system. [10], [11].

III. METHODOLOGY

A. Dataset

The experiments were conducted using the public "Water Potability" dataset downloaded from Kaggle, which contains 3,276 water samples. Each of the samples is defined by nine physicochemical attributes: pH, hardness, total dissolved solids (TDS), chloramines, sulphate, conductivity, organic carbons, trihalomethanes, and turbidity. The target feature potability specifies whether water is drinking-worthy (1) or undrinkable (0). This dataset was gathered from several open sources of water quality analysis and is frequently used as a benchmark dataset for water potability predictions based on machine learning approaches.

B. Exploratory Data Analysis

In order to validate the distribution, class imbalances, and feature correlation, EDA has been used. Fig. 1 shows the correlation between the features. From the very low value of correlations, it is clear that all the nine physical variables are independent of each other, and none of them dominates in predicting the target. Fig. 2 represents the distribution of the classes. From the figure, it is very clear that there is imbalance between the two classes, where only ~8% of data belong to potable water, and the remaining samples fall under the other class that comprises about 92%. Fig. 3 illustrates the univariate histograms of the features.

C. Data Preprocessing

The missing data in the data set were imputed through the use of the column-wise median approach to retain robustness to outliers for each feature [6]. The features were normalised using `StandardScaler` to have mean zero and unit variance. Convergence stability was ensured across all classifiers.

D. Class Imbalance Handling

Synthetic Minority Oversampling Technique (SMOTE) [3] was applied only to the training set for generating synthetic observations from the minority (drinkable) class. SMOTE generates synthetic observations through interpolation among existing minority observations, thereby not showing preferential treatment towards the majority class without any removal of existing observations.

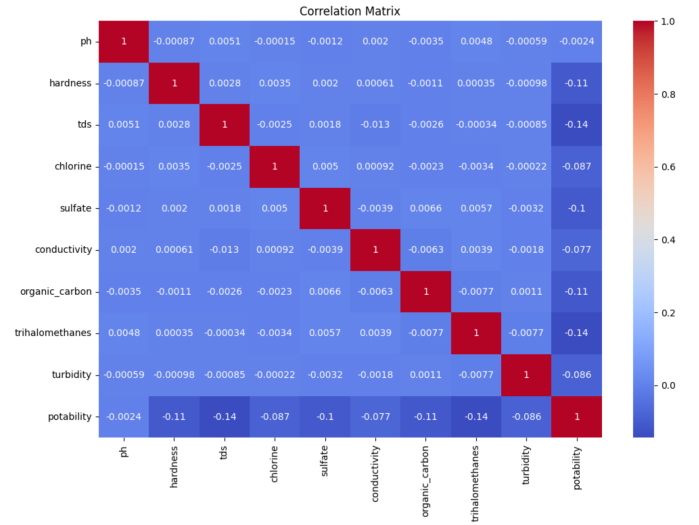


Fig. 1. Correlation matrix of physicochemical water quality parameters.

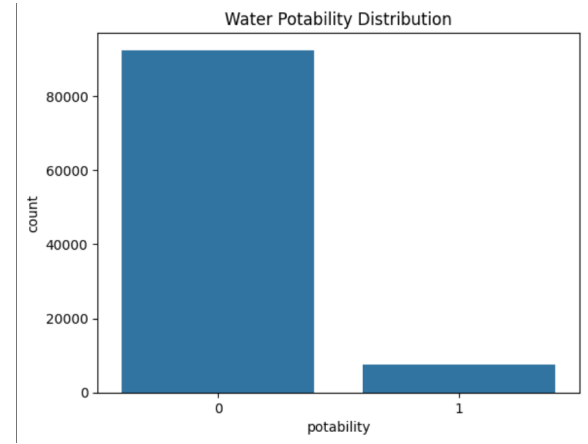


Fig. 2. Water potability class distribution showing severe class imbalance.

E. Machine Learning Models

Three supervised classifiers were implemented and evaluated [7], [8]:

- **Logistic Regression:** A linear probabilistic classifier serving as the baseline model.
- **Decision Tree:** A non-linear, rule-based classifier employing entropy-based information gain for feature splitting.
- **Random Forest:** An ensemble of decision trees trained using bootstrap aggregation (bagging) and random feature subselection, providing strong generalisation and implicit feature importance estimation.

F. SHAP Explainability

SHAP (SHapley Additive exPlanations) [9] was applied to the trained Random Forest model to provide global and instance-level interpretability. SHAP values quantify the marginal contribution of each feature to the model prediction, grounded in cooperative game-theoretic Shapley values.

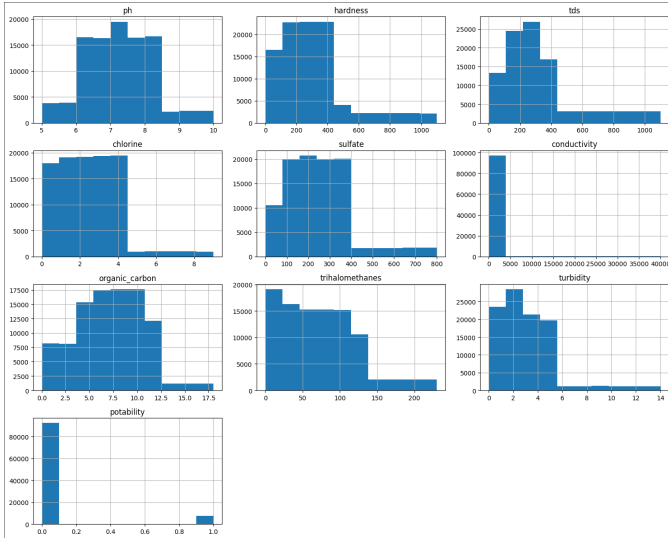


Fig. 3. Feature distributions of physicochemical parameters and target variable.

TABLE I
COMPARATIVE PERFORMANCE OF MACHINE LEARNING MODELS

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	78.28%	23.51%	82.21%	36.57%
Decision Tree	99.53%	96.79%	97.05%	96.92%
Random Forest	99.68%	96.97%	98.88%	97.92%

IV. EXPERIMENTAL SETUP

The proposed framework was implemented in Python using standard scientific computing libraries. The following libraries were utilised: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Imbalanced-learn, and SHAP.

The dataset was split into 80% training and 20% testing subsets using stratified random sampling to preserve the original class ratio in both partitions. SMOTE was applied only to the training set to prevent data leakage. Performance was evaluated using accuracy, precision, recall, F1-score, confusion matrices, and five-fold stratified cross-validation.

V. RESULTS AND DISCUSSION

A. Model Performance Comparison

Table I summarises the predictive performance of all three evaluated classifiers. The Random Forest model achieved the best overall performance with an accuracy of 99.68%, a precision of 96.97%, recall of 98.88%, and an F1-score of 97.92%. Five-fold cross-validation produced an average accuracy of 99.67%. This confirms the model's strong ability to generalize. [5].

The Logistic Regression baseline achieved 78.28% accuracy, along with a low F1-score of 36.57%. This shows how difficult it is for a linear model to capture the non-linear decision boundaries in this dataset [14]. The Decision Tree classifier greatly improved on the baseline, reaching 99.53% accuracy and an F1-score of 96.92%. [13].

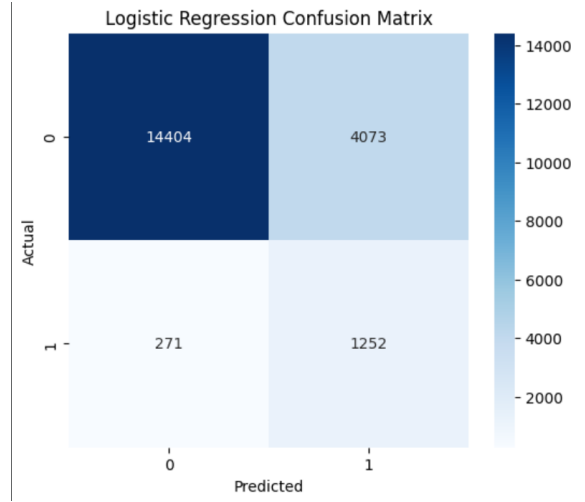


Fig. 4. Confusion matrix for the Logistic Regression classifier

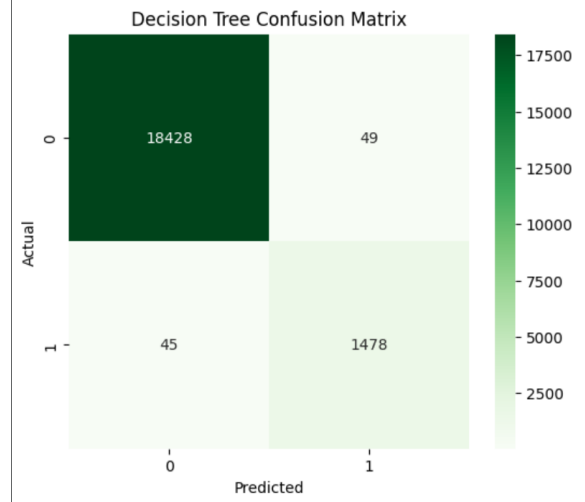


Fig. 5. Confusion matrix for the Decision Tree classifier

B. Confusion Matrices

High number of false positives (4,073) in logistic regression confusion matrix (Fig. 4) is consistent with its poor precision. The classification algorithms used for decision tree (Fig. 5) and Random Forest (Fig. 6) models exhibit significantly reduced numbers of misclassified objects, whereby random forest gives only 47 false positives and 17 false negatives. An interesting feature of random forest classification algorithm is its low number of false negatives. If such classifier were applied to solve the issue of water safety, the following considerations should be taken into account. First of all, false negative result would negatively influence the population's health, thus high recall for the potable class is required despite poor precision. For instance, the recall of random forest model with 17 false negatives out of 1,506 true positives is equal to 98.88% [16].

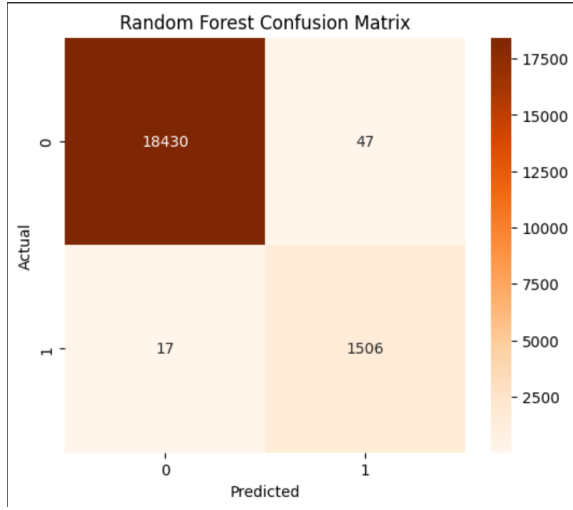


Fig. 6. Confusion matrix for the Random Forest classifier

C. Feature Importance Analysis

The following figure Fig. 7 shows the ranking of the features based on their Random Forest importance scores. The first two most influential features are conductivity ($\approx 22.2\%$) and total dissolved solids ($\approx 18.4\%$). The third is trihalomethanes ($\approx 14.6\%$), hardness ($\approx 11.3\%$), and then organic carbon ($\approx 10.9\%$). The least important feature in this case study is the pH level, which is traditionally used in water quality analysis ($\approx 3.6\%$) [10].

D. SHAP Explainability Analysis

SHAP summary plot for the random forest is shown in Fig. 8. It is observed from Fig. 8 that the global impact on the prediction of water potability depends on conductivity, TDS, and trihalomethanes. As observed from Fig. 8 using color coding (red denotes higher value and blue denotes lower value), a negative SHAP impact occurs due to high conductivity and TDS values. The results show that high values of such features decrease the likelihood of water potability [9], [11].

E. Discussion

Results of the experiments show that Random Forest as an ensemble model significantly outperformed the linear baseline of Logistic Regression as well as the one tree classifier –

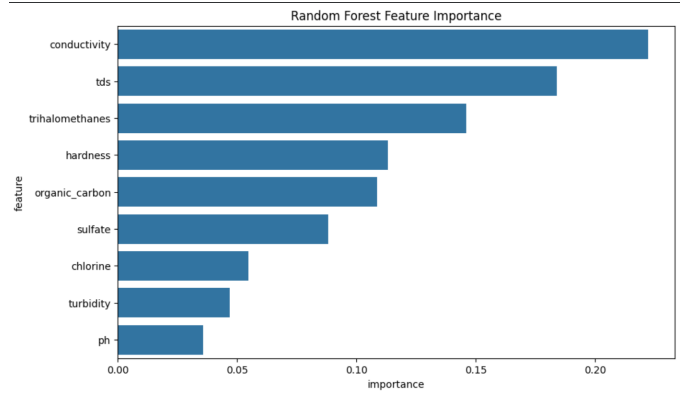


Fig. 7. Feature importance scores using the Random Forest algorithm (Conductivity and TDS are the major contributing factors)

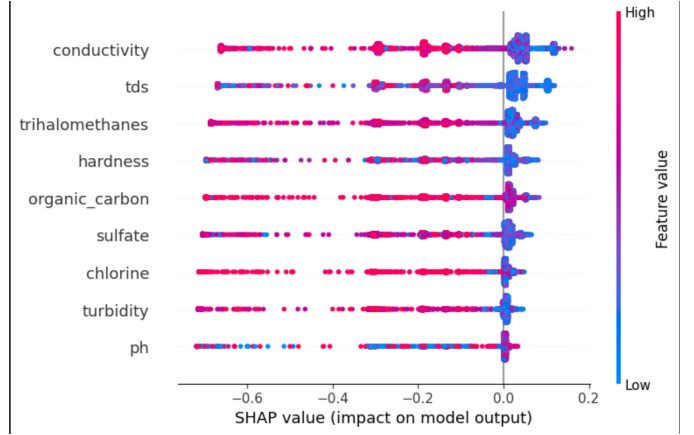


Fig. 8. Beeswarm plot using SHAP for the Random Forest algorithm (Every point is representing an individual data point and different colors represent different magnitudes of values)

Decision Tree in this case. The poor precision of Logistic Regression demonstrates that water potability classification cannot be performed using linear boundaries [14].

SMOTE was indispensable in reaching balanced recall for the rare class of water. The imbalance of the data set would make classifiers have systematic tendency to classify all samples as not potable water, causing potentially lethal amount of false negative predictions in real-life application [3].

High recall results of detecting potable water are important due to possible real-life application in which false negative predictions can cause contamination of drinkable water [4].

High predictive results from Random Forest classifier could be attributed to large amount of data available, successful imbalance resolution using SMOTE, and high ability of the ensemble model to recognize complex non-linear connections between physical and chemical parameters [7]. Five-fold cross validation was used to confirm good generalisation abilities and prevent overfitting. Although the obtained accuracy result was quite high, careful splitting and applying SMOTE only to the training sample decreased chances of data leakage and false predictions. SHAP explanations bridge the gap between

TABLE II
RANDOM FOREST FEATURE IMPORTANCE SCORES

Feature	Importance Score
Conductivity	0.2224
TDS	0.1840
Trihalomethanes	0.1462
Hardness	0.1133
Organic Carbon	0.1085
Sulfate	0.0881
Chloramines	0.0548
Turbidity	0.0470
pH	0.0357

accurate prediction and interpretability—the essential factor in the adoption of any AI-assisted systems by regulatory bodies [9]. Detection of conductivity and TDS as dominant contamination predictors helps water utilities focus their attention on particular parameters. [11].

F. Limitations

Despite the fact that the framework generated good predictions, the test used in this research is based on a publicly available benchmark dataset instead of real-time data streams [2]. Also, temporal and geographic differences in the patterns were not taken into account during modeling [4]. Further improvements may be done by using IoT-based real-time data streams and geographically varied data.

VI. CONCLUSION

The study developed an Explainable AI-driven Early Warning Framework for predicting water potability using SMOTE-balanced Random Forest classifier along with SHAP- based model interpretability. The classifier performed with 99.68% accuracy and 99.67% accuracy under cross- validation, significantly beating the baseline models of Logistic Regression and Decision Tree. Conductivity, Total Dissolved Solids (TDS), and Tri-Halomethanes emerged as the main driving factors in predicting water potability based on SHAP results. This gives insight into water resource management [9]. The suggested model solves the problem of achieving high accuracy and explainable model, both necessary conditions for deployment in intelligent water quality management system [5], [16].

Future research directions can include implementation of IoT data stream for predicting water potability in real-time, development of water quality grade prediction using multiple classes, and testing geographically diversified water quality datasets [2].

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